1 Executive summary

One frozen Perovskite 0292 acquisition | 20% Hadamard | NPL-default noise

Question. Does replacing LION's interpolation-style primal relaxation with canonical Chambolle-Pock extrapolation improve reconstruction or optimisation?

Answer. For TV, TGV-2 and TGV-3, the reconstructed images are almost identical after 8,192 iterations. The canonical update gives a lower variational objective, often substantially earlier.

Largest benefit. The objective falls much faster for TGV-2 and TGV-3. Canonical CP at 512 iterations already beats the legacy objective at 8,192 iterations for both models.

Practical limit. Faster objective convergence does not imply better PSNR or preserved details on this one case. Each PDHG iteration has nearly the same cost under either rule.

Quality and objective after 8,192 iterations

Method	Relaxation	PSNR (dB)	SSIM	HaarPSI	Objective	Time (s)
TV	LION interpolation	30.369715	0.814062	0.725788	47.61201	7.846
TV	Canonical CP	30.369713	0.814064	0.725789	47.60939	7.889
TGV-2	LION interpolation	30.436258	0.740215	0.734538	28.24963	12.174
TGV-2	Canonical CP	30.436237	0.740215	0.734535	27.84567	12.316
TGV-3	LION interpolation	30.464329	0.740222	0.733572	29.11536	16.640
TGV-3	Canonical CP	30.464373	0.740227	0.733584	27.87157	16.782

Effect size and per-iteration computational cost

Method	Objective reduction	Delta PSNR (dB)	LION ms/step	CP ms/step
TV	0.0055%	-0.0000019	0.958	0.963
TGV-2	1.4300%	-0.0000210	1.486	1.503
TGV-3	4.2719%	+0.0000439	2.031	2.049

Single-image oracle scalar weights are held fixed. All metrics are on unmodified reconstructions.

2 Experiment design and controlled variables

Same algorithms, identical parameters; only the relaxed-primal update differs

Existing LION PDHG uses interpolation: zbar[n+1] = (1-rho) z[n] + rho z[n+1]. Here rho=1, so zbar[n+1] = z[n+1].

Canonical Chambolle-Pock uses extrapolation: zbar[n+1] = z[n+1] + theta (z[n+1]-z[n]). Here theta=1, so zbar[n+1] = 2 z[n+1]-z[n].

For all three methods, the solver's initial primal and dual variables were zero; lambda/alpha weights, sampling indices, measured currents, normalisation, boundary conditions, and step sizes (tau=sigma=0.18) were unchanged. We used two local solver copies differing by one line; no LION source was modified.

PSNR-optimised scalar weights, selected previously on this image

Method	Fixed scalar weights	PDHG iterations
TV	lambda = 0.0240982484	8,192
TGV-2	alpha1 = 0.0282842712, alpha0 = 0.01	8,192
TGV-3	alpha2 = 0.032, alpha1 = 0.01, alpha0 = 0.024	8,192

Convergence recording and validation

- Nine independent zero-start runs per rule: iterations 1, 2, 64, 256, 512, 1,024, 2,048, 4,096 and 8,192. The displayed final full reconstructions and metric tables use 8,192 iterations.

- The copied legacy implementation reproduced each original archived TV, TGV-2 and TGV-3 reconstruction exactly, pixel for pixel. The two rules agree after the first iteration and differ from the second.

- Objective is the complete method-specific primal functional: one half of the squared normalized PCM data residual plus the method's fixed scalar TV/TGV penalty. Objective values are comparable between relaxation rules for the SAME method, but not across TV, TGV-2 and TGV-3.

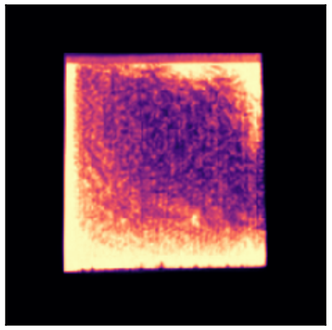
- Runtime is the elapsed time for each independent GPU run, including the solver loop and its per-run startup. Per-iteration runtime is total runtime divided by iteration count, so it is a useful estimate rather than an isolated kernel benchmark.

Provenance: original PCM frozen test data; local LION/MRPro PDHG copies; NPL-default noise.

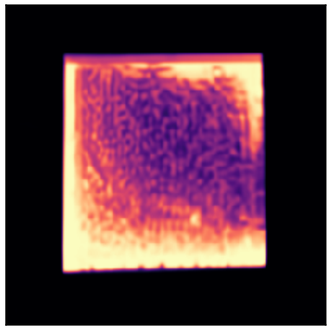
3 Full-image reconstruction comparison

GT, PnP-ADMM and UTV benchmarks plus the six relaxation-rule reconstructions

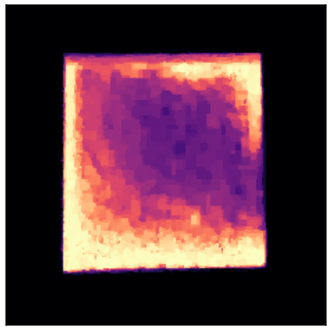
Ground truth



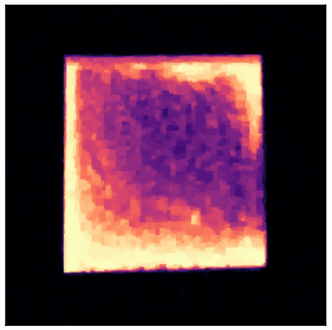
PnP-ADMM-tuned-5
PSNR 32.369354 dB



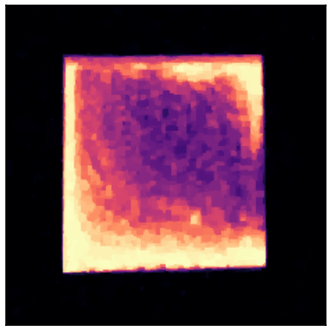
UTV C60, best rho
PSNR 29.408306 dB



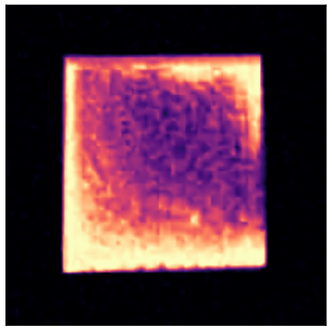
TV | LION
PSNR 30.369715 dB | SSIM 0.814062
HaarPSI 0.725788 | 8,192 steps



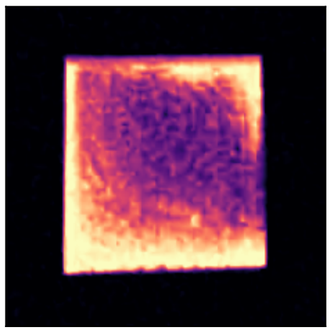
TV | Canonical CP
PSNR 30.369713 dB | SSIM 0.814064
HaarPSI 0.725789 | 8,192 steps



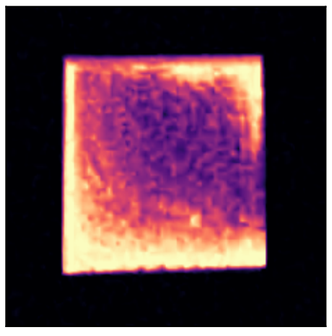
TGV-2 | LION
PSNR 30.436258 dB | SSIM 0.740215
HaarPSI 0.734538 | 8,192 steps



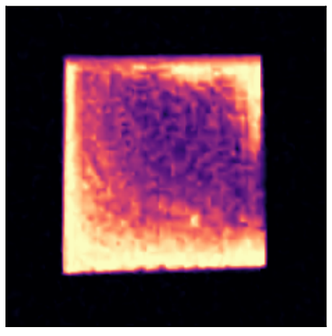
TGV-2 | Canonical CP
PSNR 30.436237 dB | SSIM 0.740215
HaarPSI 0.734535 | 8,192 steps



TGV-3 | LION
PSNR 30.464329 dB | SSIM 0.740222
HaarPSI 0.733572 | 8,192 steps



TGV-3 | Canonical CP
PSNR 30.464373 dB | SSIM 0.740227
HaarPSI 0.733584 | 8,192 steps



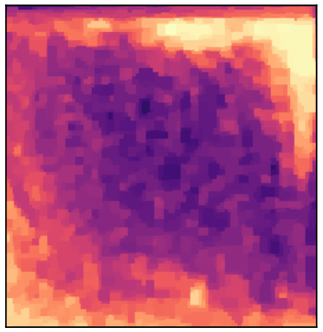
All panels use the same fixed [0,1] intensity range. Metrics use the original raw (unclipped) reconstructions.

The canonical images are ACTUAL reconstructed arrays, not placeholders or copies of the legacy panels.

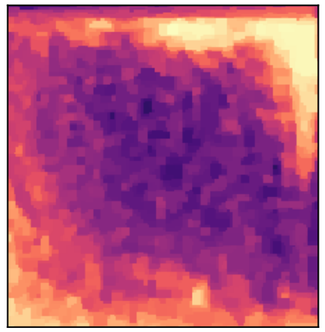
4 Matched texture crops and signed differences

Each row compares the same method under both relaxation rules

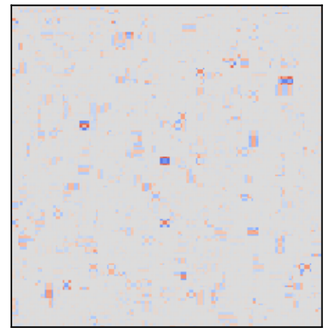
TV | LION
PSNR 30.369715 dB



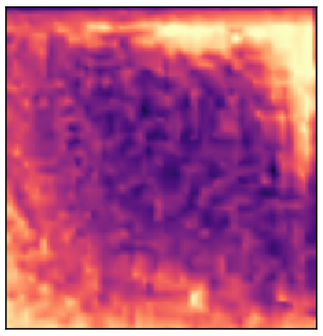
TV | Canonical CP
PSNR 30.369713 dB



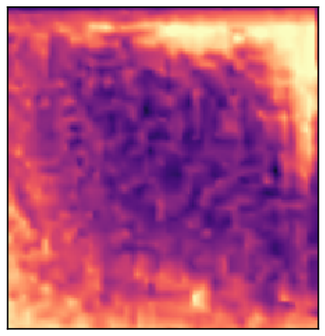
TV | CP - LION
Max |difference|: 3.263e-05



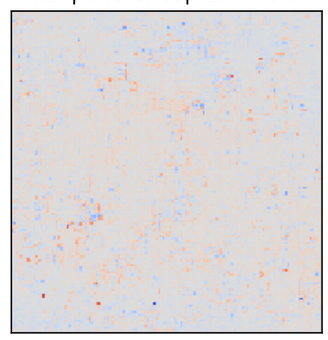
TGV-2 | LION
PSNR 30.436258 dB



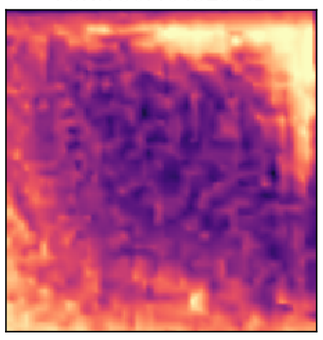
TGV-2 | Canonical CP
PSNR 30.436237 dB



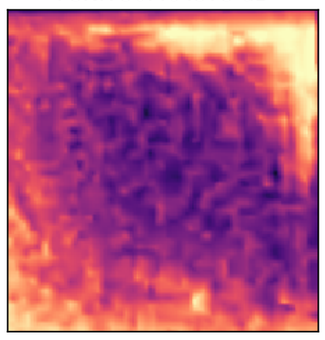
TGV-2 | CP - LION
Max |difference|: 2.747e-04



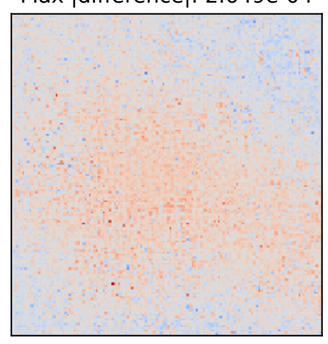
TGV-3 | LION
PSNR 30.464329 dB



TGV-3 | Canonical CP
PSNR 30.464373 dB



TGV-3 | CP - LION
Max |difference|: 2.649e-04

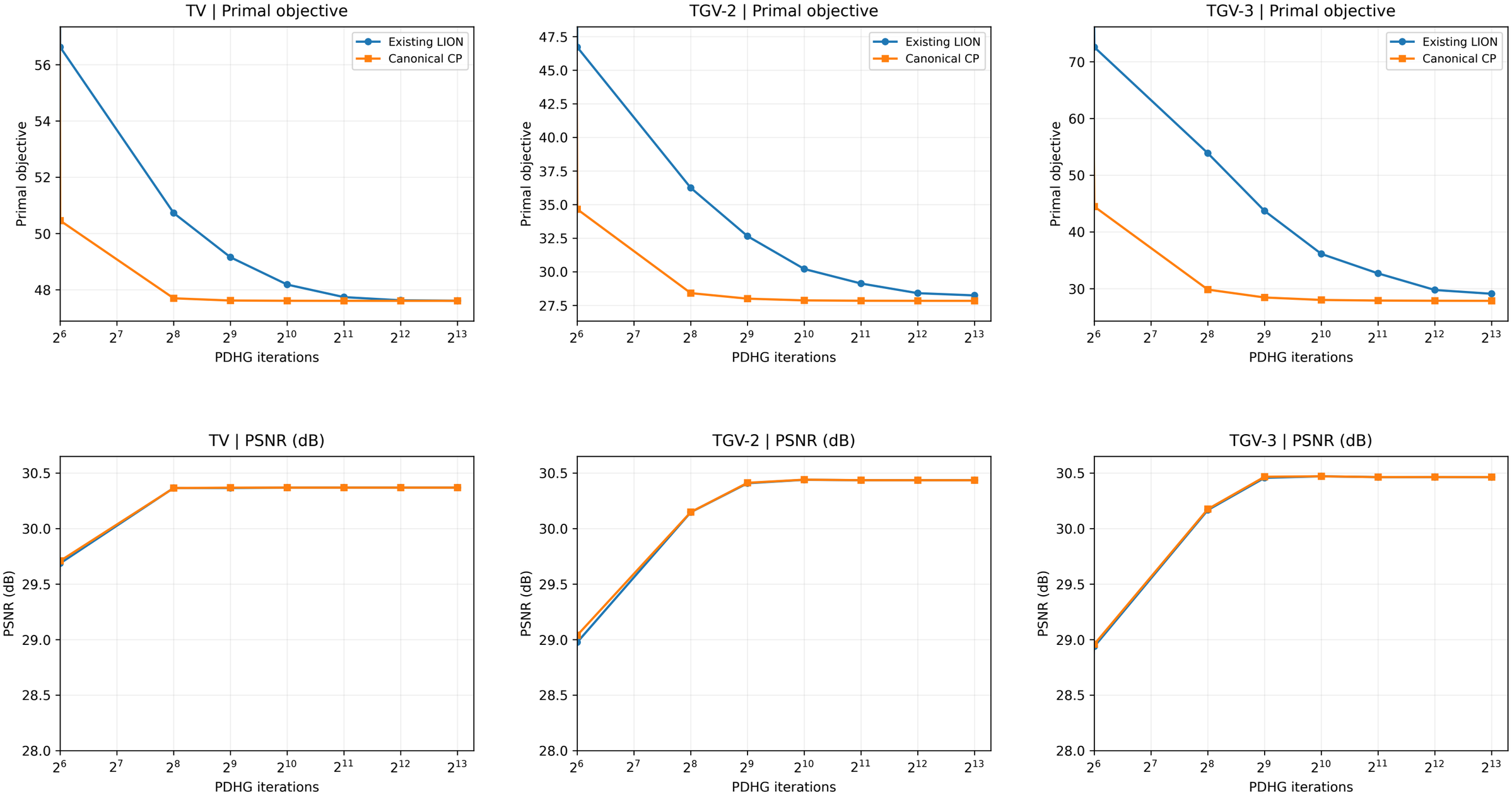


Identical crop: rows 40-184, columns 65-204. Signed differences use a separate symmetric scale for each method.

Small visual differences should not be mistaken for equivalence of the auxiliary TGV fields.

5 Objective convergence and reconstruction quality

Upper row: method-specific objectives. Lower row: PSNR at the same iteration checkpoints



The curves show independently restarted runs. Canonical CP reduces the objective much sooner, especially for TGV-2 and TGV-3.

The converged image quality need not improve even when the full variational objective decreases faster.

6 Iteration cost, savings and interpretation

Objective improvements at a fixed iteration budget versus a smaller computational budget

Method	LION 8192 s	CP 8192 s	LION ms/step	CP ms/step	CP objective @512	LION objective @8192	CP 512 s
TV	7.846	7.889	0.958	0.963	47.62022	47.61201	0.499
TGV-2	12.174	12.316	1.486	1.503	28.00785	28.24963	0.773
TGV-3	16.640	16.782	2.031	2.049	28.46180	29.11536	1.045

- Time per iteration. At 8,192 iterations, TV requires about 0.958 ms/step, TGV-2 about 1.486 ms/step, and TGV-3 about 2.031 ms/step under the existing rule. The canonical update adds almost no per-step cost.

- Relative cost. On this GPU, the TGV-2 step takes about 1.55x the TV step, while TGV-3 takes about 2.12x the TV step and 1.37x the TGV-2 step.

- Objective threshold benefit. Canonical TGV-2 and TGV-3 each reach a lower objective after 512 iterations than existing LION after 8,192. For TV, canonical CP needs 1,024 iterations to reach a lower objective than LION after 8,192. This does not certify a global optimum.

- Image quality caveat. Final PSNR changes are below 0.00005 dB in all three models. Earlier stopping under canonical CP may reduce compute time if the desired criterion is objective convergence, but should not be used without checking image quality and optimality gaps.

- Cross-method caution. The three models have different regularisers and objectives. Compare LION versus CP objectives within one method only. Compare TV/TGV-2/TGV-3 using reconstruction quality or wall-clock time, not their absolute objective magnitudes.

Primary references

Chambolle and Pock (2011), DOI 10.1007/s10851-010-0251-1; Bredies, Kunisch and Pock (2010), DOI 10.1137/090769521.

7 Checkpoint metrics and acquisition protocol

Perovskite 0292 | frozen 20% NPL-default PCM acquisition | fixed scalar regularisation weights

18 independently restarted runs = 3 methods x 2 relaxation rules x 3 iteration budgets. Each row is verified against the archived reconstruction metrics.

Steps	Method	Rule	PSNR (dB)	SSIM	HaarPSI	Objective	Data RMSE	Elapsed (s)
256	TV	LION	30.3669	0.8128	0.7274	50.729	0.04915	0.345
256	TV	CP	30.3658	0.8133	0.7261	47.699	0.04914	0.249
256	TGV-2	LION	30.1482	0.7349	0.7155	36.249	0.04666	0.388
256	TGV-2	CP	30.1494	0.7351	0.7158	28.419	0.04667	0.387
256	TGV-3	LION	30.1677	0.7312	0.7148	53.886	0.04690	0.523
256	TGV-3	CP	30.1765	0.7322	0.7155	29.852	0.04696	0.528
512	TV	LION	30.3659	0.8137	0.7263	49.157	0.04914	0.493
512	TV	CP	30.3692	0.8139	0.7258	47.620	0.04915	0.497
512	TGV-2	LION	30.4089	0.7396	0.7336	32.656	0.04669	0.767
512	TGV-2	CP	30.4130	0.7398	0.7336	28.008	0.04669	0.774
512	TGV-3	LION	30.4576	0.7385	0.7358	43.711	0.04696	1.045
512	TGV-3	CP	30.4674	0.7389	0.7361	28.462	0.04696	1.051
1,024	TV	LION	30.3692	0.8140	0.7260	48.187	0.04914	0.989
1,024	TV	CP	30.3698	0.8140	0.7258	47.611	0.04915	0.996
1,024	TGV-2	LION	30.4400	0.7402	0.7349	30.214	0.04669	1.538
1,024	TGV-2	CP	30.4412	0.7402	0.7349	27.879	0.04669	1.544
1,024	TGV-3	LION	30.4715	0.7396	0.7343	36.138	0.04693	2.089
1,024	TGV-3	CP	30.4717	0.7396	0.7343	28.029	0.04693	2.098

Definition: image residual = reconstructed image - GT. Measurement residual = A(reconstruction) - observed data.

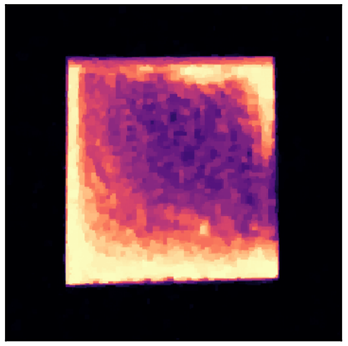
All timings were measured on tuni's RTX 4090; each run starts from zero. Step sizes tau=sigma=0.18; no weights were retuned.

Do not compare TV, TGV-2 and TGV-3 objective values with each other: they are different functionals.

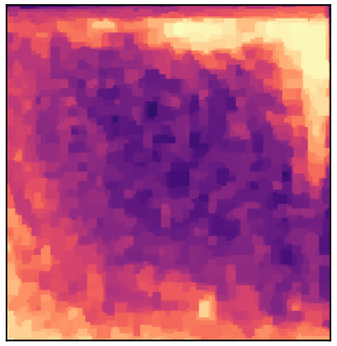
8 Reconstructions at 256 PDHG iterations

Each cell: full 256x256 image (left) and the identical texture crop (right). Same [0,1] display range.

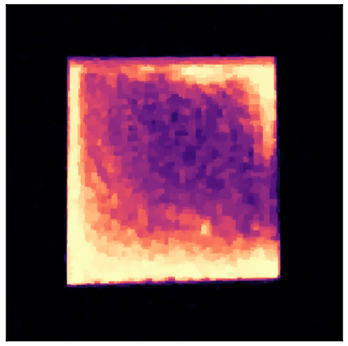
TV | Existing LION
PSNR 30.367 dB | SSIM 0.813 | HaarPSI 0.727
Elapsed 0.345 s | objective 50.729



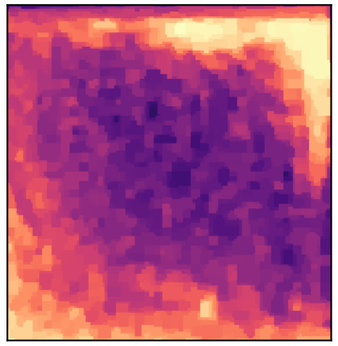
Crop: rows 40-184, cols 65-204



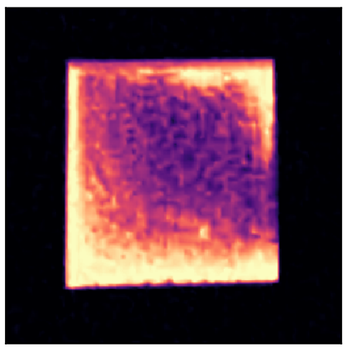
TV | Canonical CP
PSNR 30.366 dB | SSIM 0.813 | HaarPSI 0.726
Elapsed 0.249 s | objective 47.699



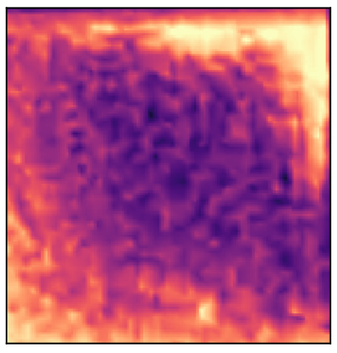
Crop: rows 40-184, cols 65-204



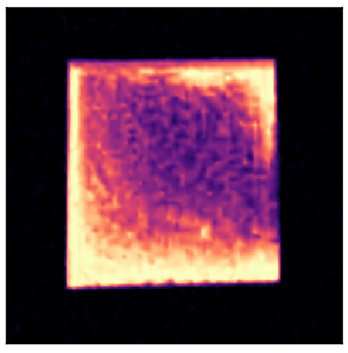
TGV-2 | Existing LION
PSNR 30.148 dB | SSIM 0.735 | HaarPSI 0.715
Elapsed 0.388 s | objective 36.249



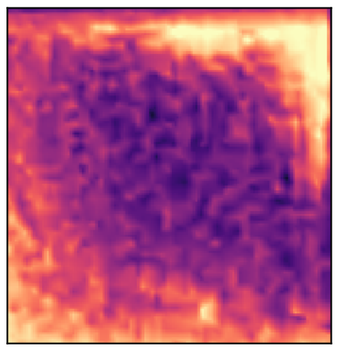
Crop: rows 40-184, cols 65-204



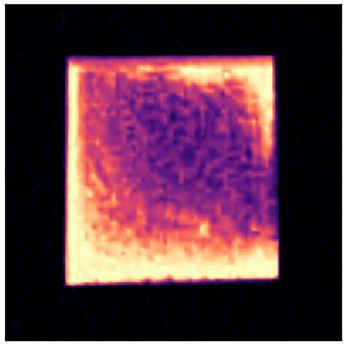
TGV-2 | Canonical CP
PSNR 30.149 dB | SSIM 0.735 | HaarPSI 0.716
Elapsed 0.387 s | objective 28.419



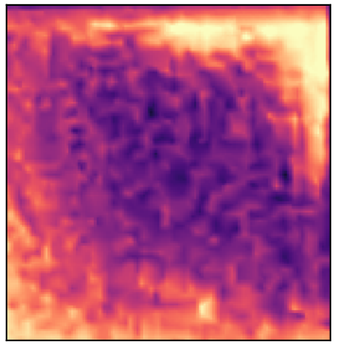
Crop: rows 40-184, cols 65-204



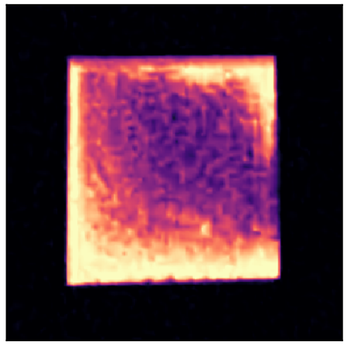
TGV-3 | Existing LION
PSNR 30.168 dB | SSIM 0.731 | HaarPSI 0.715
Elapsed 0.523 s | objective 53.886



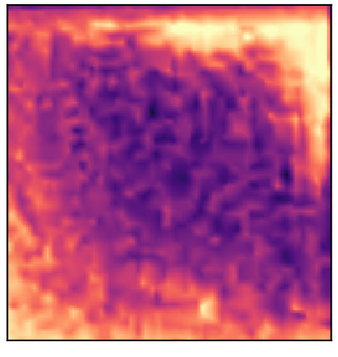
Crop: rows 40-184, cols 65-204



TGV-3 | Canonical CP
PSNR 30.176 dB | SSIM 0.732 | HaarPSI 0.715
Elapsed 0.528 s | objective 29.852



Crop: rows 40-184, cols 65-204



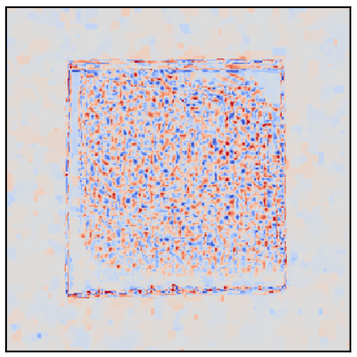
Ground-truth PSNR/SSIM/HaarPSI use raw, unclipped arrays. The same stored measurement and weights are used throughout.

The ground-truth image and 8,192-iteration comparison appear in the first six report pages.

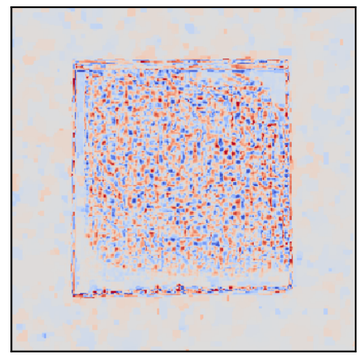
9 Signed image error at 256 iterations

Each image shows x(reconstruction) - x(ground truth). Shared symmetric colour limits across ALL methods and iteration checkpoints.

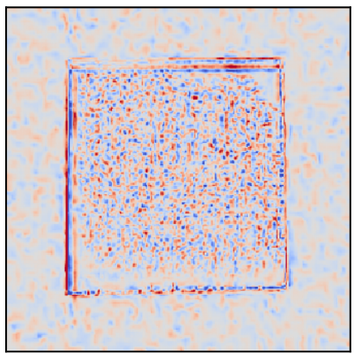
TV | Existing LION
PSNR 30.367 dB | image RMSE 0.03031



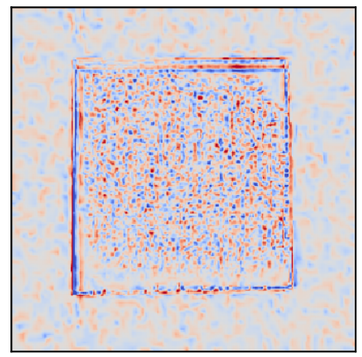
TV | Canonical CP
PSNR 30.366 dB | image RMSE 0.03032



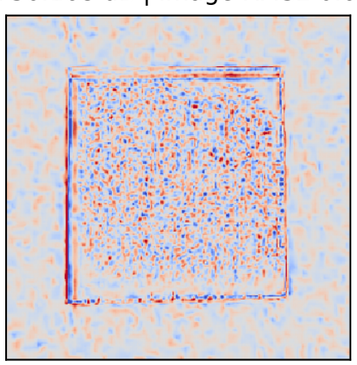
TGV-2 | Existing LION
PSNR 30.148 dB | image RMSE 0.03109



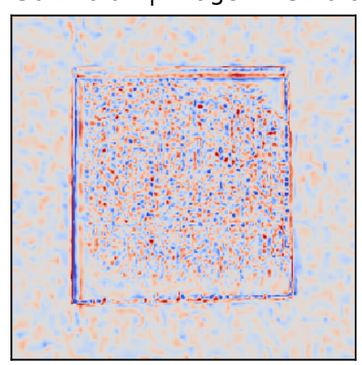
TGV-2 | Canonical CP
PSNR 30.149 dB | image RMSE 0.03108



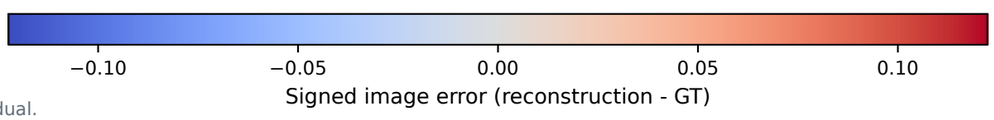
TGV-3 | Existing LION
PSNR 30.168 dB | image RMSE 0.03102



TGV-3 | Canonical CP
PSNR 30.176 dB | image RMSE 0.03099



Colour limits: +/-0.1224 (99.5th percentile pooled across all checkpoint image errors). Extreme values clip for readability.

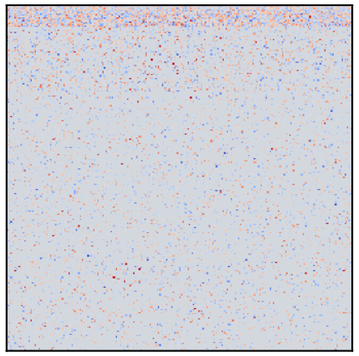


This is a spatial image-error map, not the measurement-domain data residual.

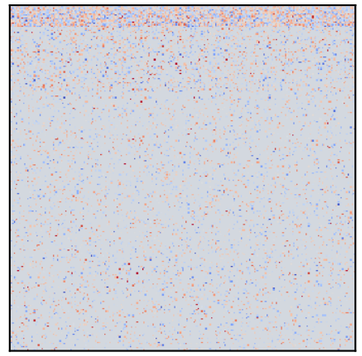
10 Measurement residuals at 256 iterations

Residual Ax-y at sampled Hadamard-pattern indices. Unmeasured indices are grey. This is NOT a spatial image.

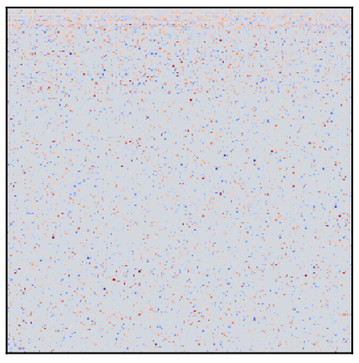
TV | Existing LION
Data residual RMSE 0.04915 | fidelity 15.835



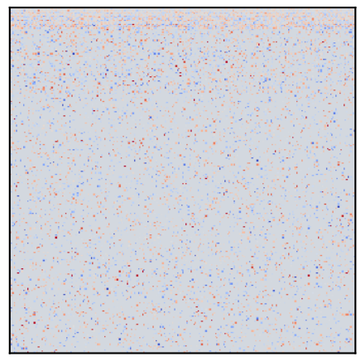
TV | Canonical CP
Data residual RMSE 0.04914 | fidelity 15.826



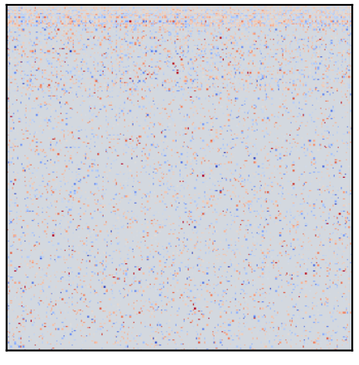
TGV-2 | Existing LION
Data residual RMSE 0.04666 | fidelity 14.268



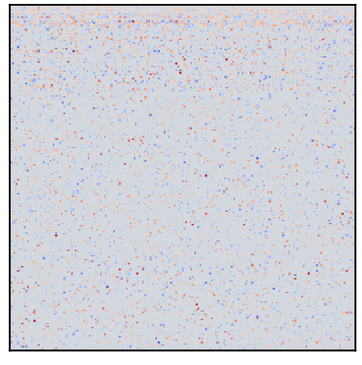
TGV-2 | Canonical CP
Data residual RMSE 0.04667 | fidelity 14.273



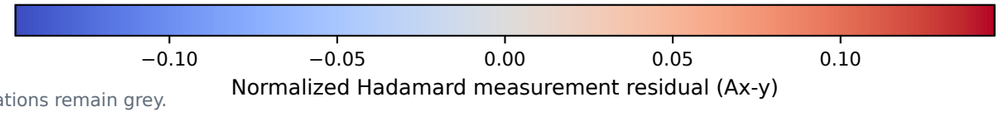
TGV-3 | Existing LION
Data residual RMSE 0.04690 | fidelity 14.417



TGV-3 | Canonical CP
Data residual RMSE 0.04696 | fidelity 14.452



Shared limits +/-0.1459 across checkpoints. Hadamard coefficient indexing does not preserve spatial pixel positions.

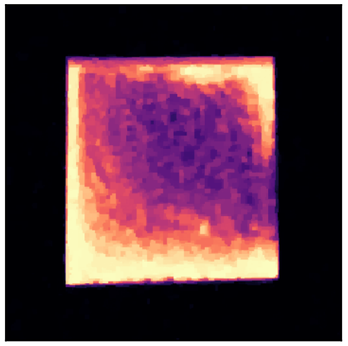


Only the 13,107 measured coefficients are coloured; 80% unsampled locations remain grey.

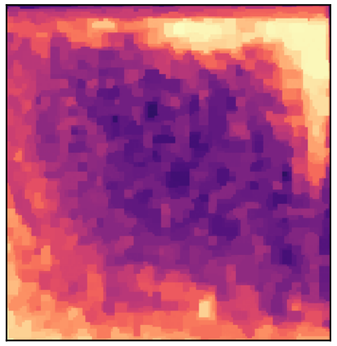
11 Reconstructions at 512 PDHG iterations

Each cell: full 256x256 image (left) and the identical texture crop (right). Same [0,1] display range.

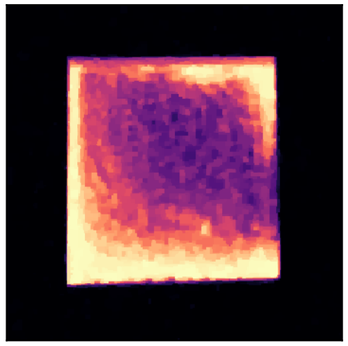
TV | Existing LION
PSNR 30.366 dB | SSIM 0.814 | HaarPSI 0.726
Elapsed 0.493 s | objective 49.157



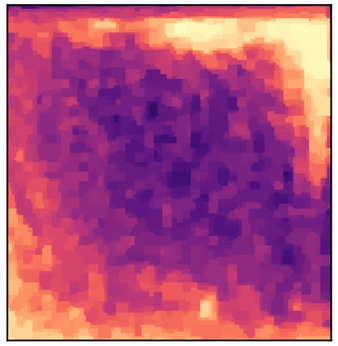
Crop: rows 40-184, cols 65-204



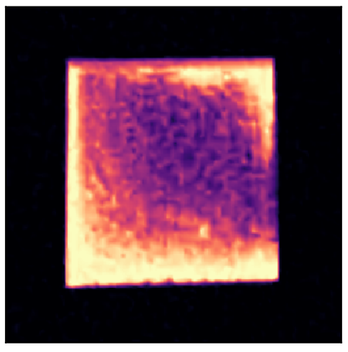
TV | Canonical CP
PSNR 30.369 dB | SSIM 0.814 | HaarPSI 0.726
Elapsed 0.497 s | objective 47.620



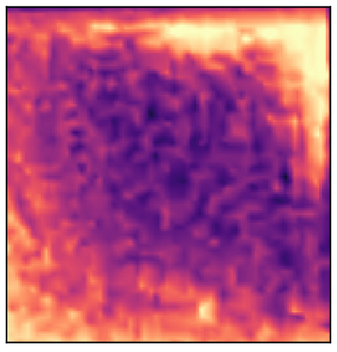
Crop: rows 40-184, cols 65-204



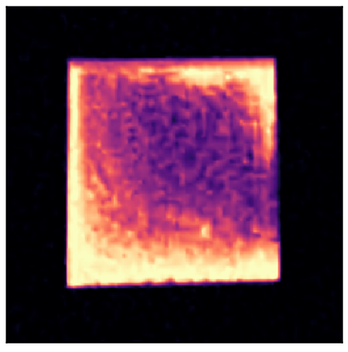
TGV-2 | Existing LION
PSNR 30.409 dB | SSIM 0.740 | HaarPSI 0.734
Elapsed 0.767 s | objective 32.656



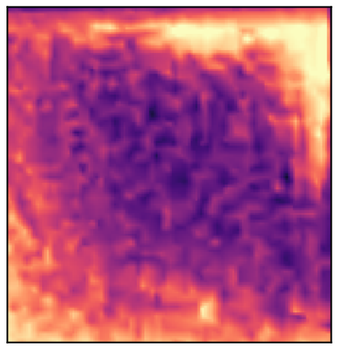
Crop: rows 40-184, cols 65-204



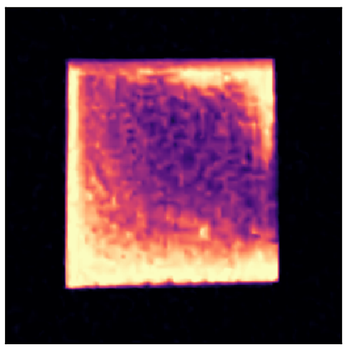
TGV-2 | Canonical CP
PSNR 30.413 dB | SSIM 0.740 | HaarPSI 0.734
Elapsed 0.774 s | objective 28.008



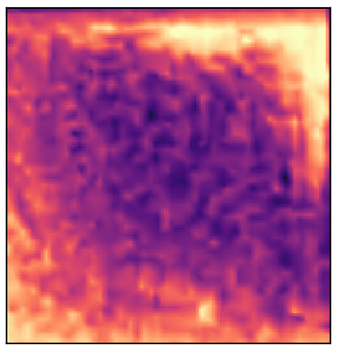
Crop: rows 40-184, cols 65-204



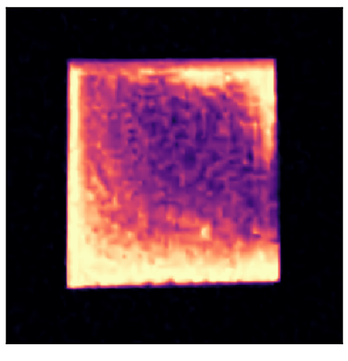
TGV-3 | Existing LION
PSNR 30.458 dB | SSIM 0.739 | HaarPSI 0.736
Elapsed 1.045 s | objective 43.711



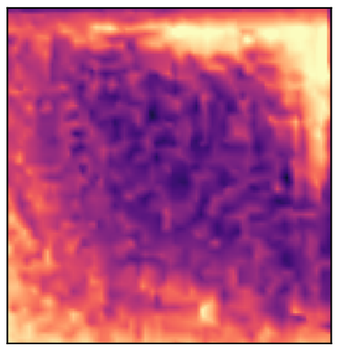
Crop: rows 40-184, cols 65-204



TGV-3 | Canonical CP
PSNR 30.467 dB | SSIM 0.739 | HaarPSI 0.736
Elapsed 1.051 s | objective 28.462



Crop: rows 40-184, cols 65-204



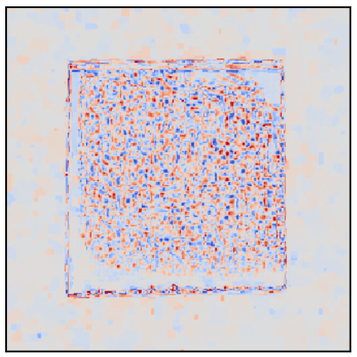
Ground-truth PSNR/SSIM/HaarPSI use raw, unclipped arrays. The same stored measurement and weights are used throughout.

The ground-truth image and 8,192-iteration comparison appear in the first six report pages.

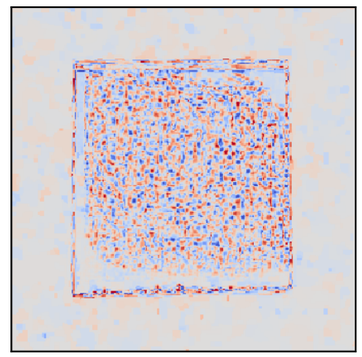
12 Signed image error at 512 iterations

Each image shows x(reconstruction) - x(ground truth). Shared symmetric colour limits across ALL methods and iteration checkpoints.

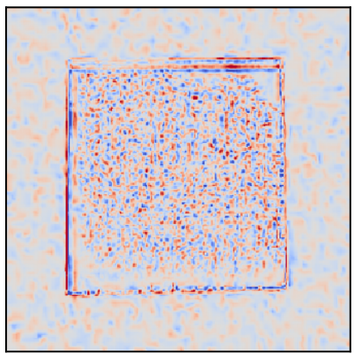
TV | Existing LION
PSNR 30.366 dB | image RMSE 0.03032



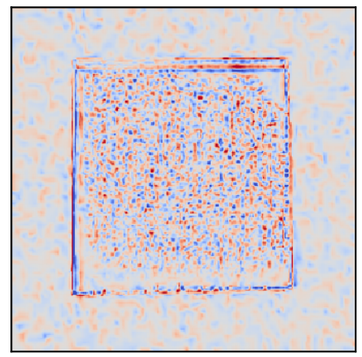
TV | Canonical CP
PSNR 30.369 dB | image RMSE 0.03031



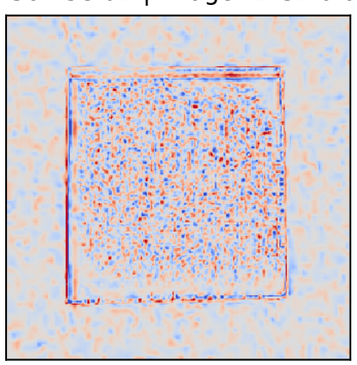
TGV-2 | Existing LION
PSNR 30.409 dB | image RMSE 0.03017



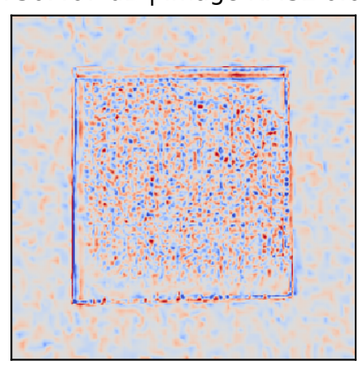
TGV-2 | Canonical CP
PSNR 30.413 dB | image RMSE 0.03015



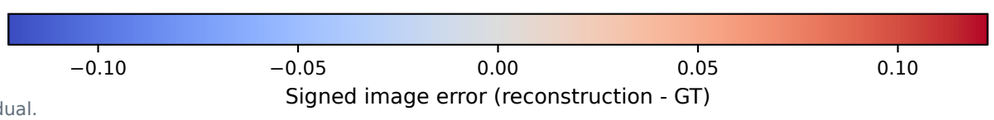
TGV-3 | Existing LION
PSNR 30.458 dB | image RMSE 0.03000



TGV-3 | Canonical CP
PSNR 30.467 dB | image RMSE 0.02997



Colour limits: +/-0.1224 (99.5th percentile pooled across all checkpoint image errors). Extreme values clip for readability.

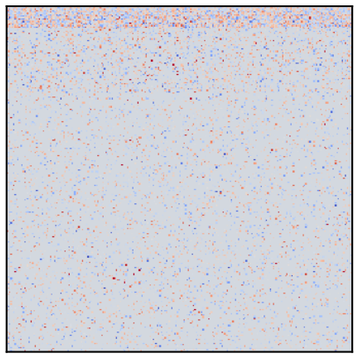


This is a spatial image-error map, not the measurement-domain data residual.

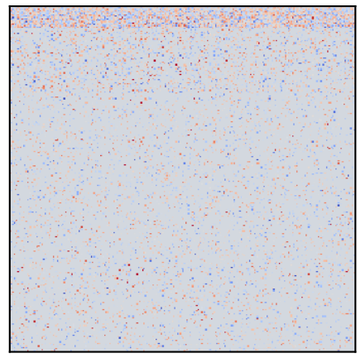
13 Measurement residuals at 512 iterations

Residual Ax-y at sampled Hadamard-pattern indices. Unmeasured indices are grey. This is NOT a spatial image.

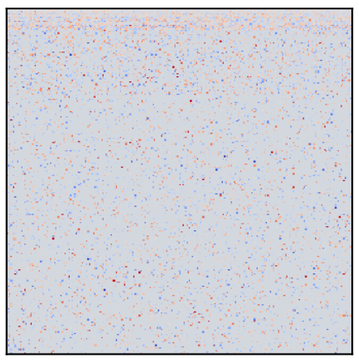
TV | Existing LION
Data residual RMSE 0.04914 | fidelity 15.826



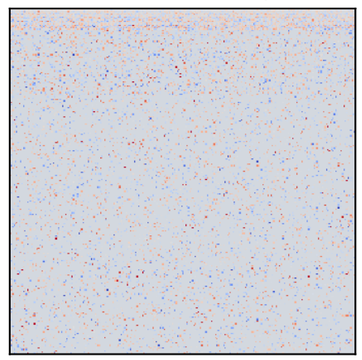
TV | Canonical CP
Data residual RMSE 0.04915 | fidelity 15.829



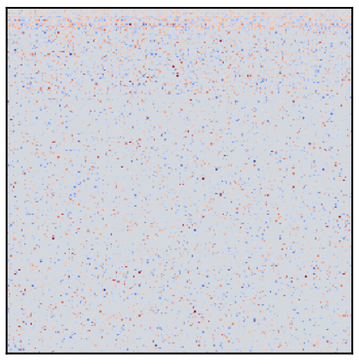
TGV-2 | Existing LION
Data residual RMSE 0.04669 | fidelity 14.286



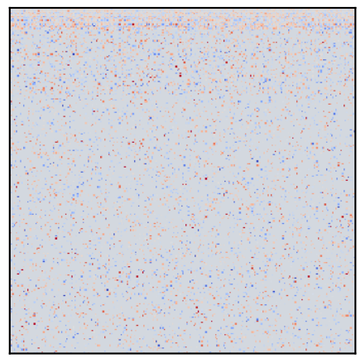
TGV-2 | Canonical CP
Data residual RMSE 0.04669 | fidelity 14.283



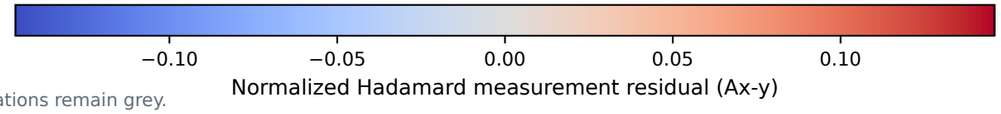
TGV-3 | Existing LION
Data residual RMSE 0.04696 | fidelity 14.452



TGV-3 | Canonical CP
Data residual RMSE 0.04696 | fidelity 14.455



Shared limits +/-0.1459 across checkpoints. Hadamard coefficient indexing does not preserve spatial pixel positions.

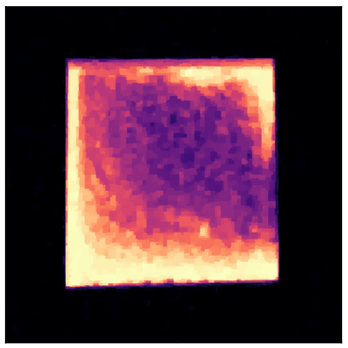


Only the 13,107 measured coefficients are coloured; 80% unsampled locations remain grey.

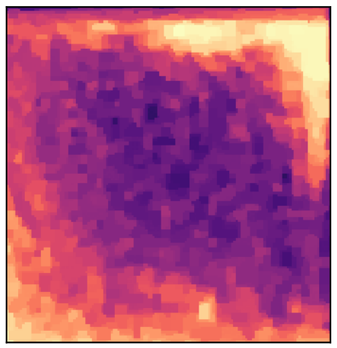
14 Reconstructions at 1,024 PDHG iterations

Each cell: full 256x256 image (left) and the identical texture crop (right). Same [0,1] display range.

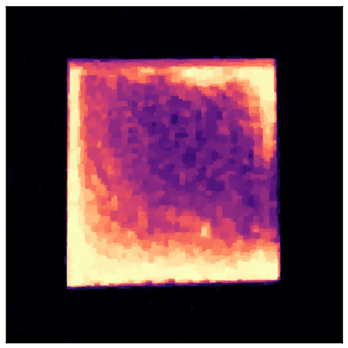
TV | Existing LION
PSNR 30.369 dB | SSIM 0.814 | HaarPSI 0.726
Elapsed 0.989 s | objective 48.187



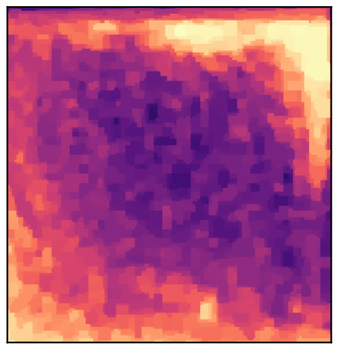
Crop: rows 40-184, cols 65-204



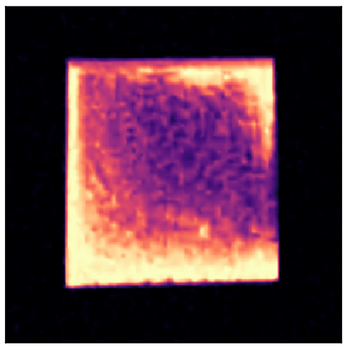
TV | Canonical CP
PSNR 30.370 dB | SSIM 0.814 | HaarPSI 0.726
Elapsed 0.996 s | objective 47.611



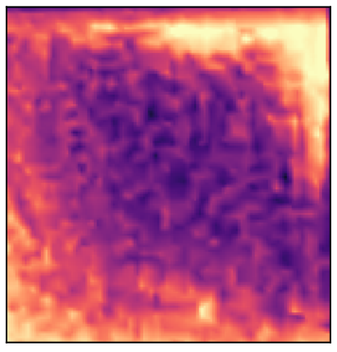
Crop: rows 40-184, cols 65-204



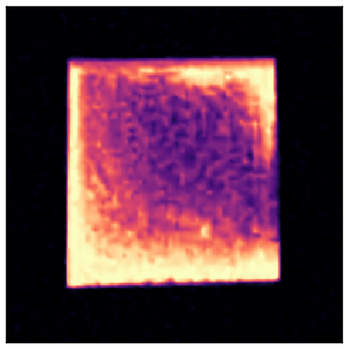
TGV-2 | Existing LION
PSNR 30.440 dB | SSIM 0.740 | HaarPSI 0.735
Elapsed 1.538 s | objective 30.214



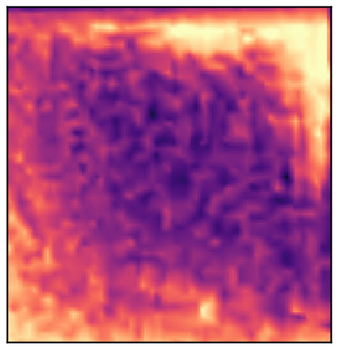
Crop: rows 40-184, cols 65-204



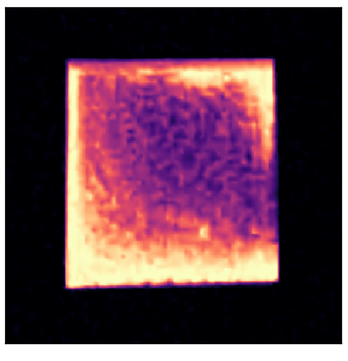
TGV-2 | Canonical CP
PSNR 30.441 dB | SSIM 0.740 | HaarPSI 0.735
Elapsed 1.544 s | objective 27.879



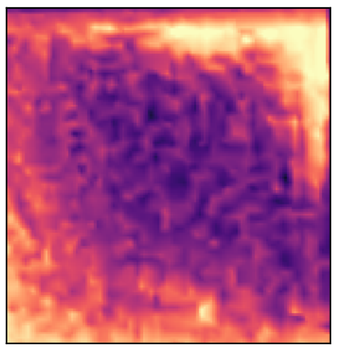
Crop: rows 40-184, cols 65-204



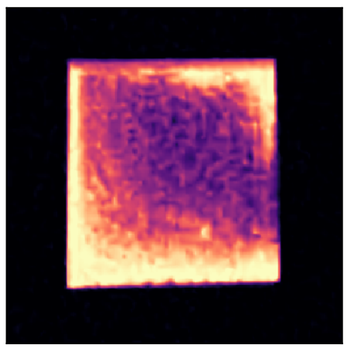
TGV-3 | Existing LION
PSNR 30.472 dB | SSIM 0.740 | HaarPSI 0.734
Elapsed 2.089 s | objective 36.138



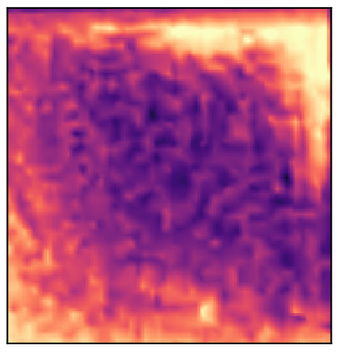
Crop: rows 40-184, cols 65-204



TGV-3 | Canonical CP
PSNR 30.472 dB | SSIM 0.740 | HaarPSI 0.734
Elapsed 2.098 s | objective 28.029



Crop: rows 40-184, cols 65-204



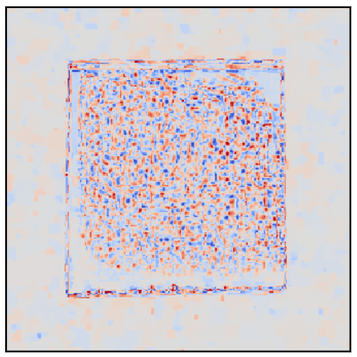
Ground-truth PSNR/SSIM/HaarPSI use raw, unclipped arrays. The same stored measurement and weights are used throughout.

The ground-truth image and 8,192-iteration comparison appear in the first six report pages.

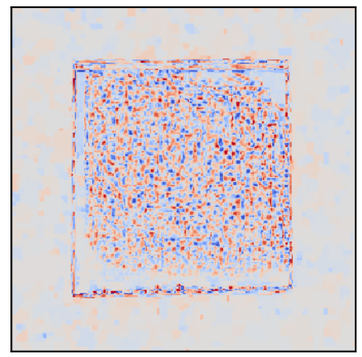
15 Signed image error at 1,024 iterations

Each image shows x(reconstruction) - x(ground truth). Shared symmetric colour limits across ALL methods and iteration checkpoints.

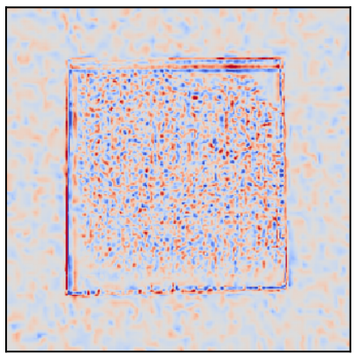
TV | Existing LION
PSNR 30.369 dB | image RMSE 0.03031



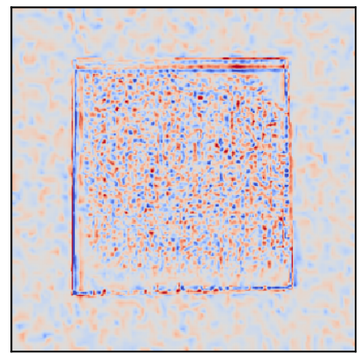
TV | Canonical CP
PSNR 30.370 dB | image RMSE 0.03030



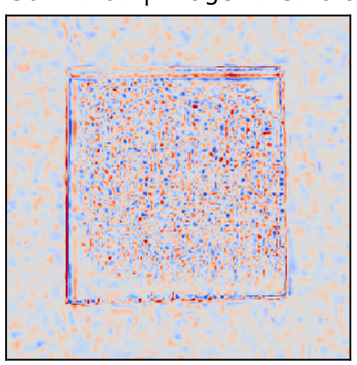
TGV-2 | Existing LION
PSNR 30.440 dB | image RMSE 0.03006



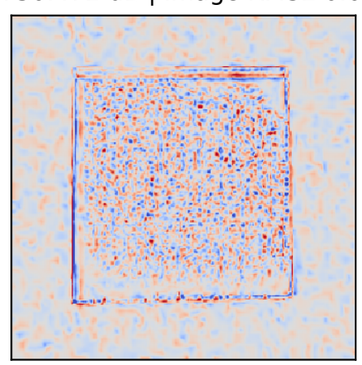
TGV-2 | Canonical CP
PSNR 30.441 dB | image RMSE 0.03006



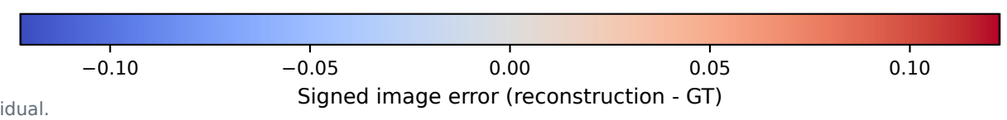
TGV-3 | Existing LION
PSNR 30.472 dB | image RMSE 0.02995



TGV-3 | Canonical CP
PSNR 30.472 dB | image RMSE 0.02995



Colour limits: +/-0.1224 (99.5th percentile pooled across all checkpoint image errors). Extreme values clip for readability.

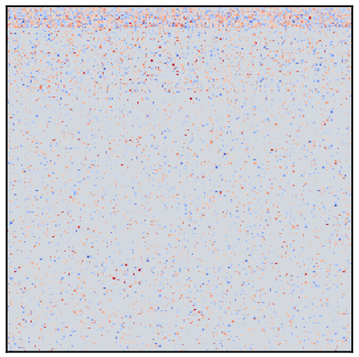


This is a spatial image-error map, not the measurement-domain data residual.

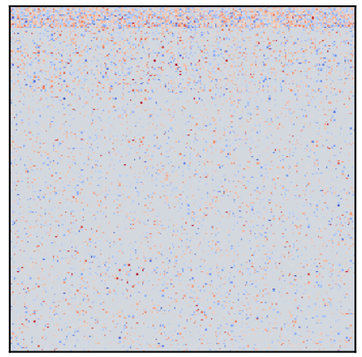
16 Measurement residuals at 1,024 iterations

Residual Ax-y at sampled Hadamard-pattern indices. Unmeasured indices are grey. This is NOT a spatial image.

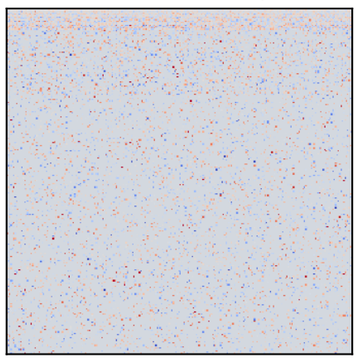
TV | Existing LION
Data residual RMSE 0.04914 | fidelity 15.828



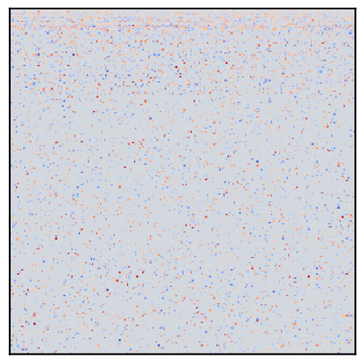
TV | Canonical CP
Data residual RMSE 0.04915 | fidelity 15.829



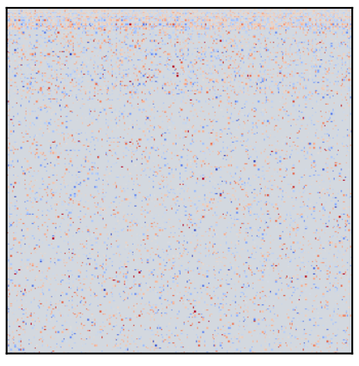
TGV-2 | Existing LION
Data residual RMSE 0.04669 | fidelity 14.285



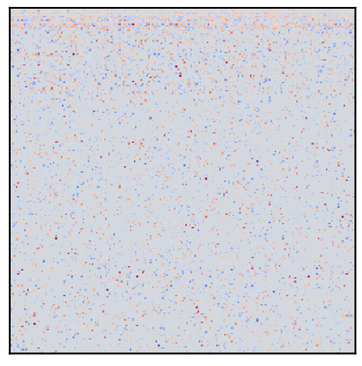
TGV-2 | Canonical CP
Data residual RMSE 0.04669 | fidelity 14.284



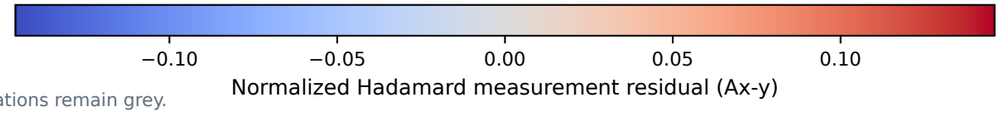
TGV-3 | Existing LION
Data residual RMSE 0.04693 | fidelity 14.435



TGV-3 | Canonical CP
Data residual RMSE 0.04693 | fidelity 14.434



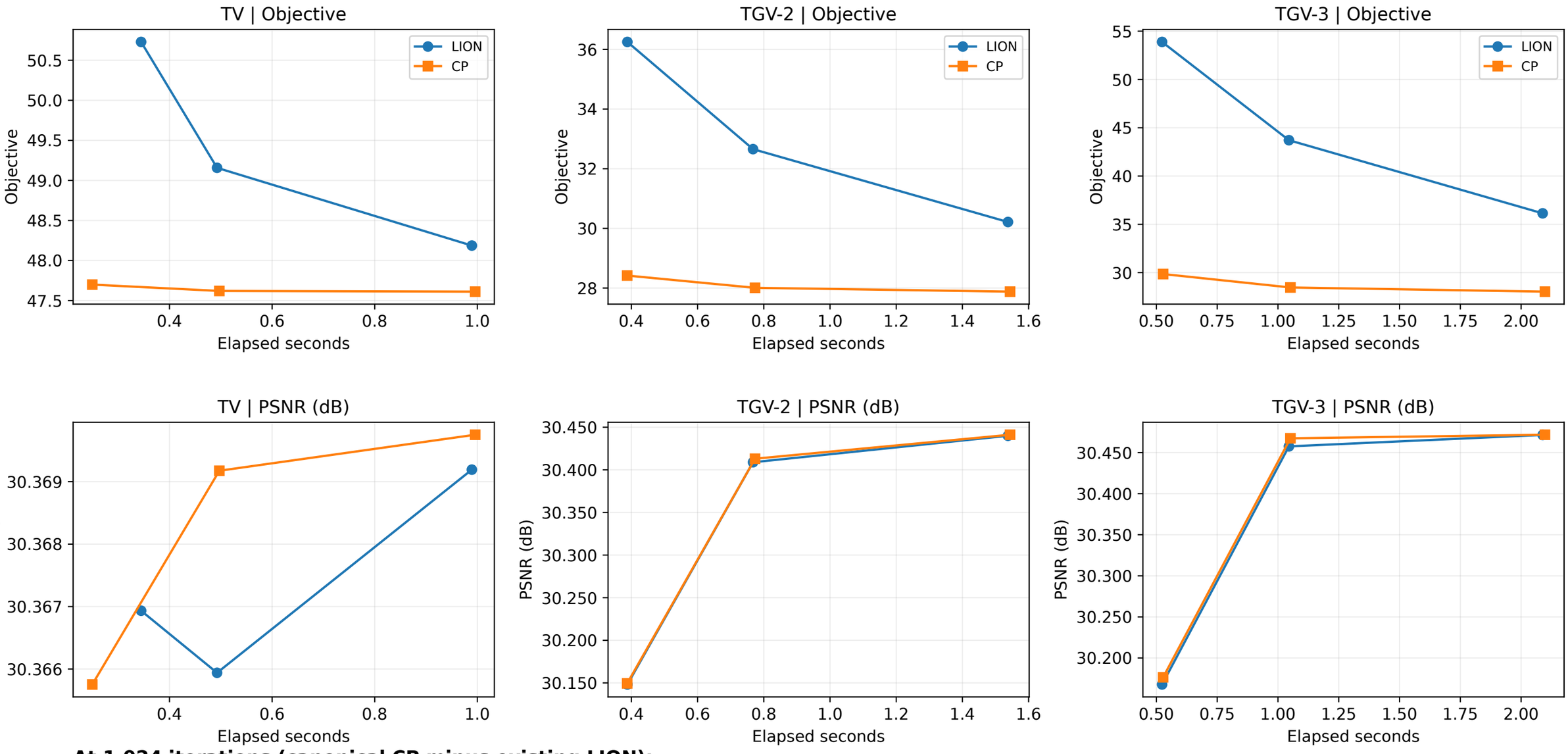
Shared limits +/-0.1459 across checkpoints. Hadamard coefficient indexing does not preserve spatial pixel positions.



Only the 13,107 measured coefficients are coloured; 80% unsampled locations remain grey.

17 Runtime and convergence implications

Observed checkpoint timings on one GPU, with both relaxation rules and fixed regularisation parameters



At 1,024 iterations (canonical CP minus existing LION):

TV	LION 0.966 ms/step	CP 0.972 ms/step	objective change -0.57512	PSNR change +0.000559 dB
TGV-2	LION 1.502 ms/step	CP 1.508 ms/step	objective change -2.33459	PSNR change +0.001213 dB
TGV-3	LION 2.040 ms/step	CP 2.049 ms/step	objective change -8.10819	PSNR change +0.000223 dB

Canonical CP changes the convergence trajectory rather than the cost of a PDHG iteration. Objective thresholds and image quality must be assessed separately.

Single GT-tuned case. Runtime is noisy for very short runs; compare at matched iteration budgets.